A connected graph with integer diminished Sombor energy

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Abstract

We construct a connected simple bipartite graph on 77 vertices with diminished Sombor energy 72, disproving Movahedi's nonintegrality conjecture. Its six nonzero eigenvalues are $\pm 10,\ \pm(13+\sqrt{29}),\ \pm(13-\sqrt{29})$. The construction also gives connected counterexamples of arbitrarily large minimum degree.

Task 1. The conjecture and its resolution

For a finite simple graph G, let d_v be the degree of v. Its diminished Sombor matrix and energy are

$$(M_{\mathrm{DS}}(G))_{uv} = \begin{cases} \dfrac{\sqrt{d_u^2+d_v^2}}{d_u+d_v}, & uv \in E(G), \\ 0, & uv \notin E(G), \end{cases} \qquad \mathcal{E}_{\mathrm{DS}}(G) = \sum_{\lambda \in \operatorname{Spec} M_{\mathrm{DS}}(G)} |\lambda|,$$

where eigenvalues are counted with multiplicity. Conjecture 5.1 of [1], p. 19, asserts that this energy is never an integer. This is the question recorded as MathDB 369710. The following counterexample is connected and has minimum degree 21.

Theorem 1. *There is a connected simple bipartite graph G with 77 vertices, 980 edges, 28 vertices of degree 21, and 49 vertices of degree 28, such that*

$$\operatorname{Spec} M_{\mathrm{DS}}(G) = \{10, -10,\ 13+\sqrt{29}, -13-\sqrt{29},\ 13-\sqrt{29}, -13+\sqrt{29},\ 0^{(71)}\}.$$

In particular, $\mathcal{E}_{\mathrm{DS}}(G) = 72$.

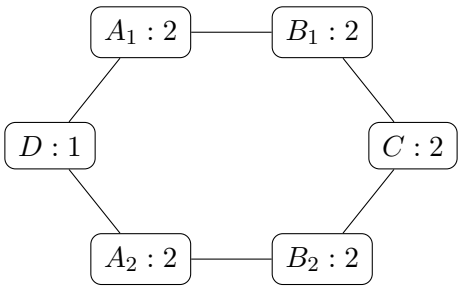
An eleven-vertex graph. Define H using disjoint independent vertex classes

$$|A_1| = |A_2| = |B_1| = |B_2| = |C| = 2, \qquad |D| = 1.$$

Put in every edge between each of the following six pairs of classes, and put in no other edges:

$$A_1B_1, \quad A_2B_2, \quad A_1D, \quad A_2D, \quad CB_1, \quad CB_2.$$

Thus H is the blowup of a six-cycle with five classes of size two and one class of size one. Its bipartition is $(A_1 \cup A_2 \cup C,\ B_1 \cup B_2 \cup D)$. It is connected, has 20 edges, and has degree 3 on $A_1 \cup A_2$ and degree 4 on all other vertices.



Each drawn edge denotes all edges between its two independent classes.

Exact spectrum of the base graph

The diminished Sombor weights on the degree pairs $(3,4)$ and $(4,4)$ are, respectively,

$$u = \frac{5}{7}, \qquad w = \frac{1}{\sqrt{2}}.$$

Order the two sides of the bipartition as above, and write $M_{\mathrm{DS}}(H) = \left(\begin{smallmatrix} 0 & B \\ B^{\mathsf{T}} & 0 \end{smallmatrix}\right)$. If J_2 is the two-by-two all-ones matrix and $\mathbf{1}_2 = (1,1)^{\mathsf{T}}$, then

$$B = \begin{pmatrix} uJ_2 & 0 & u\mathbf{1}_2 \\ 0 & uJ_2 & u\mathbf{1}_2 \\ wJ_2 & wJ_2 & 0 \end{pmatrix}.$$

Vectors summing to zero inside a vertex class are annihilated by the appropriate block. On the subspaces of vectors constant on classes, use the normalized indicator vectors of the classes. The resulting three-by-three matrix is

$$R = \begin{pmatrix} 10/7 & 0 & 5\sqrt{2}/7 \\ 0 & 10/7 & 5\sqrt{2}/7 \\ \sqrt{2} & \sqrt{2} & 0 \end{pmatrix}.$$

Consequently the nonzero singular values of B are those of R. Replacing the first two coordinates on both sides by their normalized difference and sum gives the orthogonal block decomposition

$$R \;\sim\; \frac{10}{7} \;\oplus\; T, \qquad T = \begin{pmatrix} 10/7 & 10/7 \\ 2 & 0 \end{pmatrix}.$$

Let $s_+ \geq s_- > 0$ be the singular values of T. The trace and determinant identities give

$$s_+^2 + s_-^2 = \operatorname{tr}(TT^{\mathsf{T}}) = \frac{396}{49}, \qquad s_+ s_- = |\det T| = \frac{20}{7}.$$

It follows that

$$(s_+ + s_-)^2 = \frac{676}{49}, \qquad s_+ + s_- = \frac{26}{7}, \qquad s_\pm = \frac{13 \pm \sqrt{29}}{7}.$$

Both values are positive because $13^2 > 29$. Since B has rank three, the standard symmetric bipartite block form gives

$$\operatorname{Spec} M_{\mathrm{DS}}(H) = \left\{ \pm\frac{10}{7},\ \pm\frac{13+\sqrt{29}}{7},\ \pm\frac{13-\sqrt{29}}{7},\ 0^{(5)} \right\}.$$

In particular,

$$\mathcal{E}_{\mathrm{DS}}(H) = 2\left(\frac{10}{7} + \frac{26}{7}\right) = \frac{72}{7}.$$

Blowup and integer energy

Lemma 1. *For a positive integer r, replace each vertex of a simple graph F by r independent vertices, and each edge by all r^2 cross edges. Denote the resulting graph by $F^{(r)}$. Then*

$$M_{\mathrm{DS}}(F^{(r)}) = M_{\mathrm{DS}}(F) \otimes J_r, \qquad \mathcal{E}_{\mathrm{DS}}(F^{(r)}) = r\,\mathcal{E}_{\mathrm{DS}}(F).$$

Proof. Each new vertex corresponding to v has degree rd_v. On an edge the weight is unchanged, since

$$\frac{\sqrt{(rd_u)^2+(rd_v)^2}}{rd_u+rd_v}=\frac{\sqrt{d_u^2+d_v^2}}{d_u+d_v}.$$

This proves the matrix identity. The eigenvalues of J_r are r once and zero $r-1$ times. The tensor-product eigenvalues now give the energy identity. □

Proof of Theorem 1. Take $G = H^{(7)}$. Equivalently, in the six-class construction replace each class of size two by a class of size 14, and the singleton class by a class of size 7. The graph is simple and bipartite. It is connected: every path in H lifts between corresponding classes, and two vertices in the same class have a common neighbor. There are $7 \cdot 11 = 77$ vertices and $7^2 \cdot 20 = 980$ edges. The $4 \cdot 7 = 28$ vertices over $A_1 \cup A_2$ have degree 21; the other $7 \cdot 7 = 49$ vertices have degree 28. Lemma 1 multiplies the six nonzero eigenvalues of H by seven and supplies the other 71 zero eigenvalues. Finally,

$$\mathcal{E}_{\mathrm{DS}}(G) = 2\big(10 + (13+\sqrt{29}) + (13-\sqrt{29})\big) = 72.$$

□

More generally, $\mathcal{E}_{\mathrm{DS}}(H^{(r)}) = 72r/7$, which is an integer exactly when $7 \mid r$. Thus $H^{(7t)}$, for every integer $t \geq 1$, is a connected counterexample with $77t$ vertices, $980t^2$ edges, minimum degree $21t$, and energy $72t$.

Reproducible verification. The accompanying standard-library script independently constructs the eleven- and seventy-seven-vertex graphs from their edges, checks the degree-based weights in exact arithmetic, and verifies the spectrum by matrix and Gram-matrix characteristic-polynomial calculations. For G,

$$\det(xI - M_{\mathrm{DS}}(G)) = x^{71}(x^2-100)(x^4-396x^2+19600).$$

These finite checks supplement the symbolic proof above.

Four follow-up questions

Task 2. Determine the minimum order of a connected simple graph with positive integer diminished Sombor energy.

Task 3. Decide whether every tree with at least one edge has irrational diminished Sombor energy.

Task 4. Decide whether a simple graph can have odd integer diminished Sombor energy.

Task 5. Construct, or rule out, a connected graph with positive integer diminished Sombor energy in which no two distinct vertices have the same open neighborhood.

References

[1] F. Movahedi, *Diminished Sombor matrix, spectral radius, and energy of the graphs*, arXiv:2508.06531v1 (2025), 22 pp., arxiv.org/abs/2508.06531. Definition: p. 2; Conjecture 5.1: p. 19.